

Descriptive Analytics Report: Netflix

Catalog Growth, Genre Mix & Content Composition at Scale

Source dataset: netflix_titles.csv

Report type: Descriptive Analytics

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Tracking Content Additions, Genre Mix, Ratings, Regional Demand & Catalog Composition

Note on data:

This report uses the supplied Netflix titles dataset containing 8,807 records spanning 127 countries and 42 distinct genre tags. The dataset is a catalog snapshot (one row per title, with a "date added to Netflix" field) rather than a raw user-activity or viewership event stream. Therefore the analysis describes what the supplied catalog records show — title counts, genres, countries, ratings and durations — and does not assume access to viewership, streaming-minutes, or subscriber-level data.

1. Objective

Build a descriptive analytics layer over the Netflix titles catalog to answer the following recurring content-strategy questions:

- How is the Netflix catalog growing in terms of content additions over time?
- How does the release-year mix of catalog titles trend over time?
- What is the split between Movies and TV Shows?
- Which genres contribute the highest number of titles?
- Which genre tags dominate the overall catalog?
- Which countries produce the highest number of titles?
- How is content demand distributed geographically?
- What is the distribution of content ratings (maturity ratings)?
- How long are movies, typically, and how many seasons do TV shows run?
- Which directors contribute the most titles to the catalog?
- How does genre diversity relate to a country's total title count?
- Are there weekday vs weekend patterns in when content is added?
- How "fresh" is the catalog — how long after release do titles typically arrive on Netflix?

2. How This Would Work at Real Scale

Processing a large, continuously-updated content catalog is an engineering problem before it is an analytics problem. A realistic architecture would ingest title metadata as it is added or updated, aggregate it by time, genre, country and rating, store both raw and processed data, and expose the resulting KPIs through a reporting or data-visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the titles dataset	Google, Claude & ChatGPT
Batch processing	Aggregate by genre, country and time	Python (pandas) / Jupyter Notebook
Storage	Retain raw and processed data	CSV files / a data warehouse
Aggregation	Catalog KPIs and summary metrics	Jupyter Notebook / Python
Report	Present analysis, graphs and insights	Data visualization / MS Word

3. Content Type Split — Movies vs TV Shows

Across the supplied 8,807 catalog records, Movies account for 69.62% of titles (6,131 titles) and TV Shows account for 30.38% (2,676 titles). The chart below shows this split.

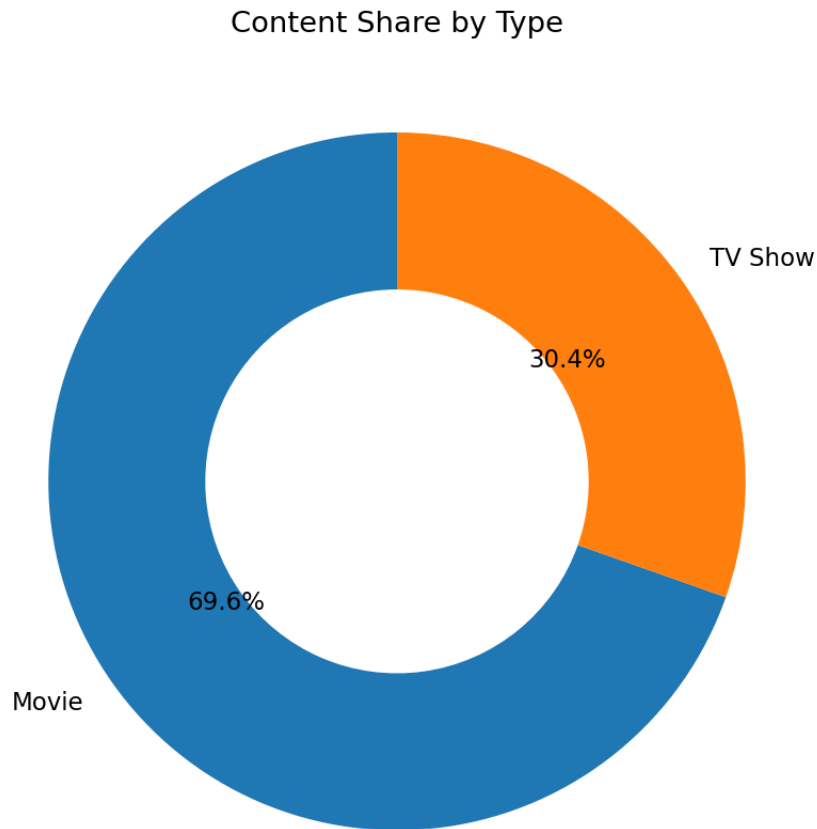


Figure 1. Content share by type

INSIGHTS:

- Movies make up the clear majority of the catalog, at roughly two out of every three titles.
- TV Shows form a substantial minority, and typically generate repeat engagement across seasons.
- The type split is a useful baseline before drilling into genre, country, or rating breakdowns.

KPI:

Total titles: 8,807

Movies: 6,131 (69.62%)

TV Shows: 2,676 (30.38%)

4. Monthly Content Additions Over Time

The dataset records a "date added" for 8,797 titles, spanning 2008-01-01 to 2021-09-25. The chart below shows how many titles were added to the catalog each month across that window.

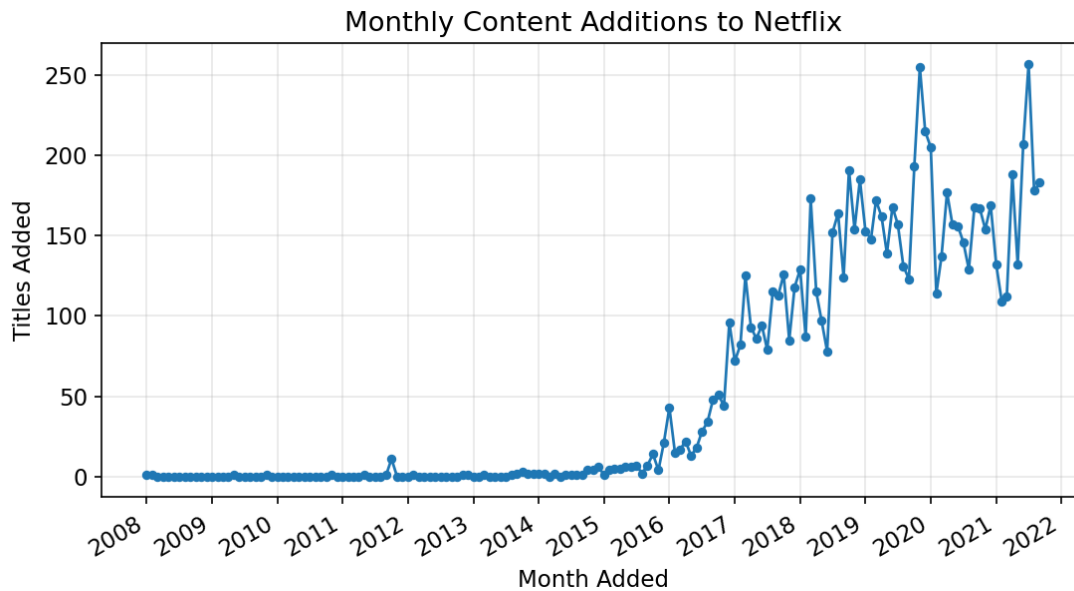


Figure 2. Monthly content additions

INSIGHTS:

- Monthly additions were minimal before roughly 2015 and accelerated sharply from 2016 onward.
- Peak months indicate periods of aggressive catalog expansion.
- Monitoring monthly additions can help track the pace of catalog growth over time.
- This trend can support content-acquisition and licensing planning.

KPI:

Peak month: 2021-07 (257 titles)

Average titles added per month: 53.3

Date range covered: 2008-01-01 to 2021-09-25

5. Yearly Content Additions Trend

Aggregating additions by calendar year gives a clearer view of year-over-year catalog growth, smoothing out month-to-month noise.

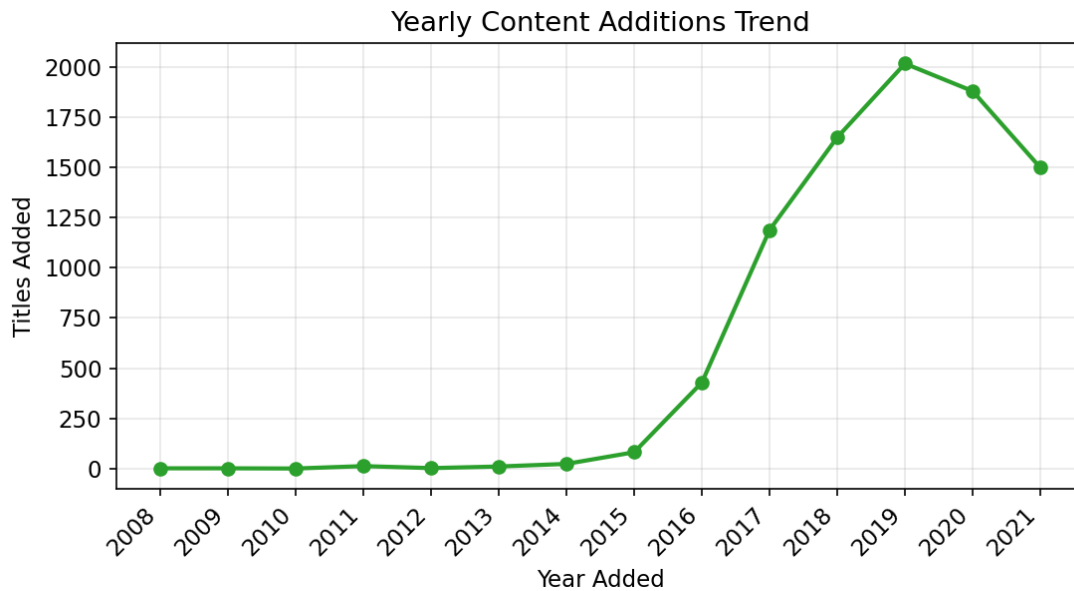


Figure 3. Yearly content additions trend

INSIGHTS:

- Additions grew every year from 2015 through 2019, consistent with a period of rapid catalog expansion.
- 2020 and 2021 show a slight pull-back versus the 2019 peak in the supplied data.
- Year-level trends are useful for high-level catalog-growth reporting to stakeholders.

KPI:

Highest addition year: 2019 (2,016 titles)

Years covered: 2008–2021

6. Top Genres by Title Count

Genre tags are multi-valued in this dataset (a title can carry several genre labels), producing 19,323 genre-tag occurrences across 42 distinct genres. The chart below ranks the ten most common genre tags.

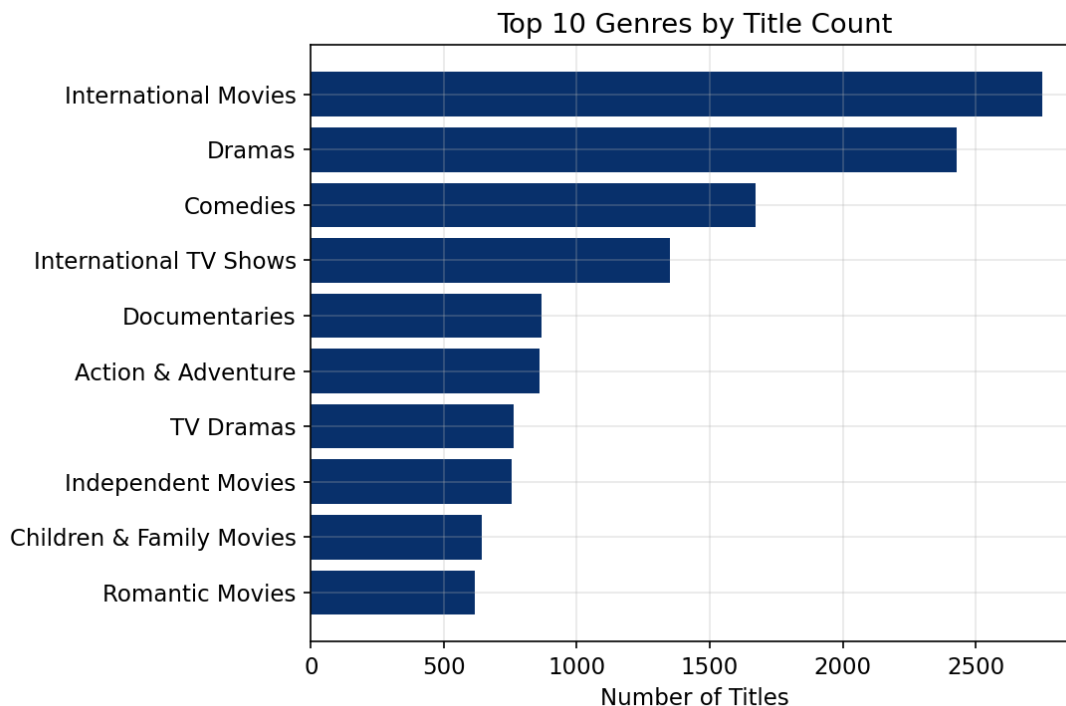


Figure 4. Top 10 genres by title count

INSIGHTS:

- "International Movies" is the single most common genre tag, appearing on 2,752 titles.
- Drama and Comedy tags also feature heavily, reflecting broad mainstream appeal.
- Because genres are multi-valued, these counts overlap — a title can contribute to more than one bar.

KPI:

Top genre: International Movies

Titles tagged: 2,752

Share of all titles: 31.25%

7. Genre Tag Share of the Catalog

Looking at genre-tag share rather than raw counts shows how concentrated the catalog is in a handful of dominant genres versus a long tail of niche categories.

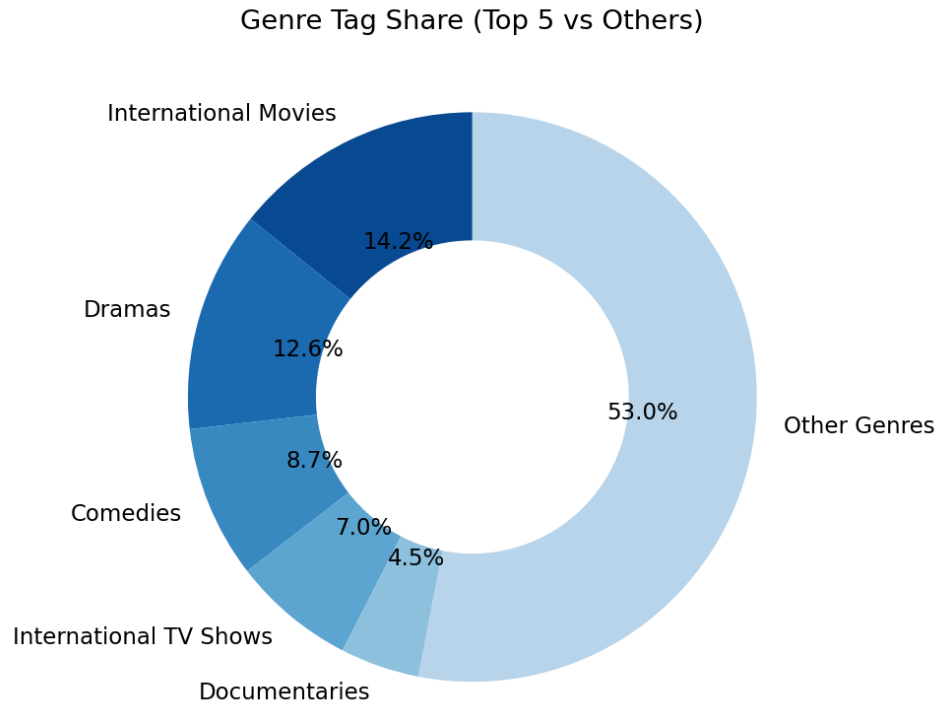


Figure 5. Genre tag share — top 5 vs. others

INSIGHTS:

- The top five genres account for a substantial share of all genre-tag occurrences.
- A long tail of niche genres (Anime, Stand-Up Comedy, Classic Movies, etc.) makes up the remainder.
- This concentration can guide which genres merit dedicated curation or promotion.

KPI:

Genres represented: 42

Total genre-tag occurrences: 19,323

8. Countries Producing the Highest Title Counts

Using each title's primary (first-listed) country of production, United States is the largest single contributor, with 3,211 titles in the supplied dataset. The distribution shows that catalog origin is concentrated in a subset of countries rather than evenly spread.

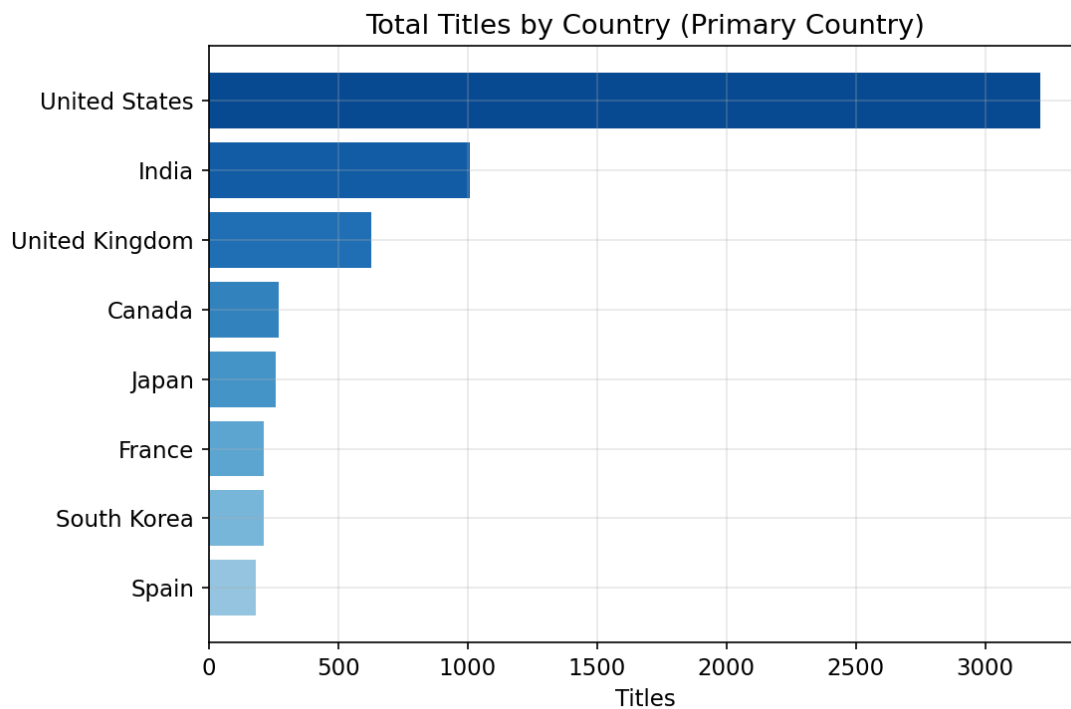


Figure 6. Total titles by primary country

INSIGHTS:

- Catalog origin is concentrated in a handful of countries rather than evenly distributed.
- The United States and India together account for a large share of total titles.
- Country-level breakdowns can support regional content-licensing and localization decisions.

KPI:

Top-performing country: United States

Recorded titles: 3,689

Share of all titles: 41.89%

9. Regional Demand — Titles by Country

Title production is distributed across 127 countries in the supplied dataset. The chart below re-presents the top eight countries with exact counts labelled for easier reference.

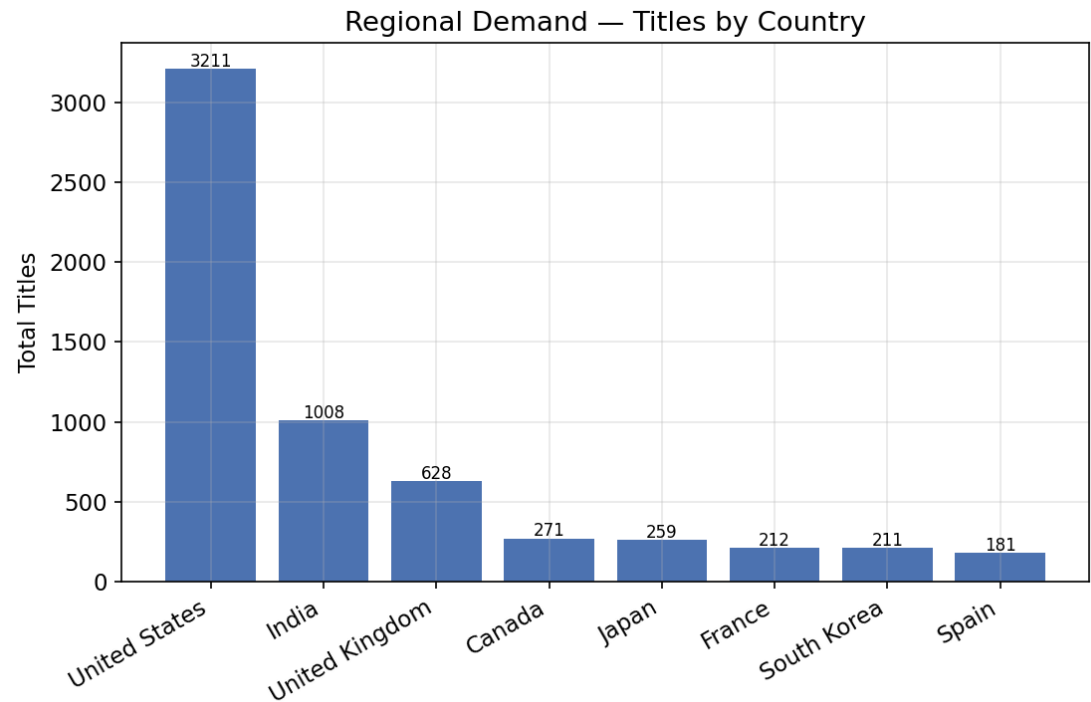


Figure 7. Regional demand — titles by country

INSIGHTS:

- Demand (as measured by title count) is distributed unevenly across countries.
- United States represents the strongest single market in the supplied data.
- Country-level demand information can support catalog-expansion and market-entry decisions.

KPI:

Countries covered (min. 5 titles in scatter set): 54

Total unique countries referenced: 127

10. Content Rating Distribution

The average content-rating mix skews toward mature audiences: 45.54% of rated titles carry a mature rating (TV-MA, R, or NC-17), versus 23.31% carrying a family-friendly rating (G, PG, TV-G, TV-Y, TV-Y7, TV-PG).

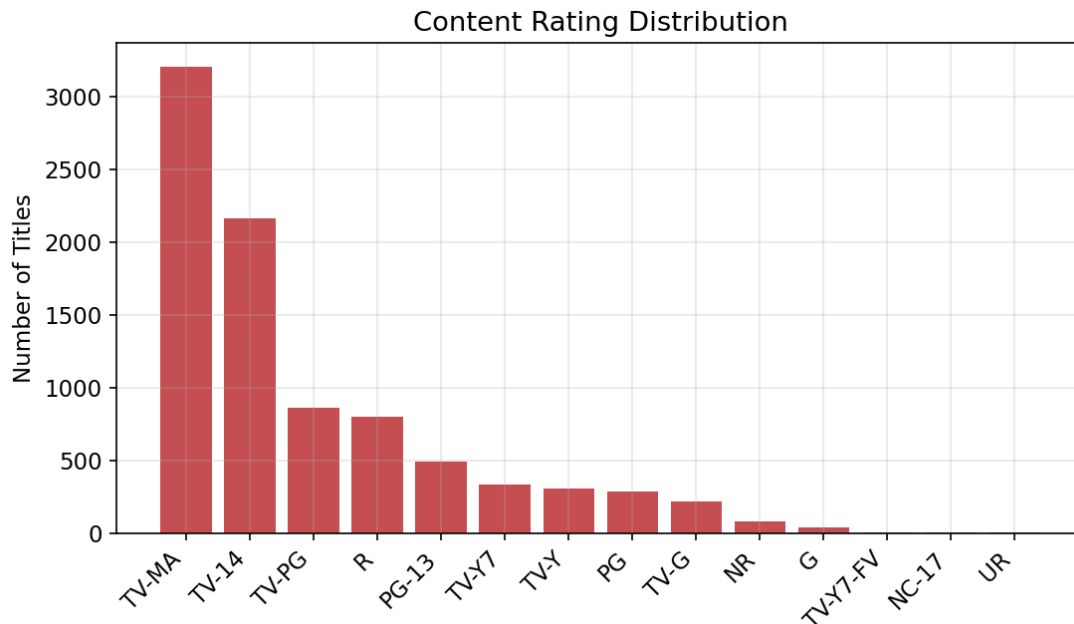


Figure 8. Content rating distribution

INSIGHTS:

- "TV-MA" is the single most common rating, covering 36.43% of rated titles.
- Mature-audience ratings outnumber family-friendly ratings in the supplied catalog.
- Rating mix is a useful input for parental-control and content-discovery features.

KPI:

Most common rating: TV-MA

Titles with that rating: 3,207

Mature-audience share: 45.54%

Family-friendly share: 23.31%

11. Movie Duration Distribution

Across 6,131 movies with a parsable runtime, the average movie duration is 99.6 minutes, ranging from 3 to 312 minutes.

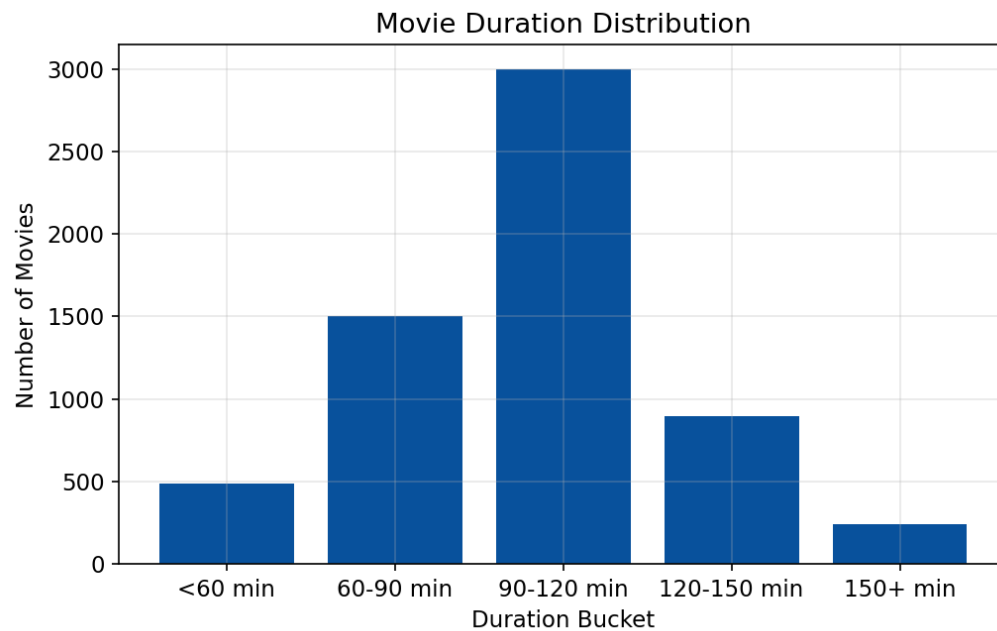


Figure 9. Movie duration distribution

INSIGHTS:

- Most movies fall between 90 and 120 minutes, the conventional feature-length range.
- A smaller number of titles run under 60 minutes or over 150 minutes — likely documentaries, specials, or epics.
- Duration mix can inform recommendations for time-constrained viewing sessions.

KPI:

Average movie duration: 99.6 minutes

Shortest movie: 3 minutes

Longest movie: 312 minutes

12. TV Show Season-Count Distribution

Across 2,676 TV shows, the average show runs for 1.76 seasons, and 67% of shows have only a single season in the supplied data.

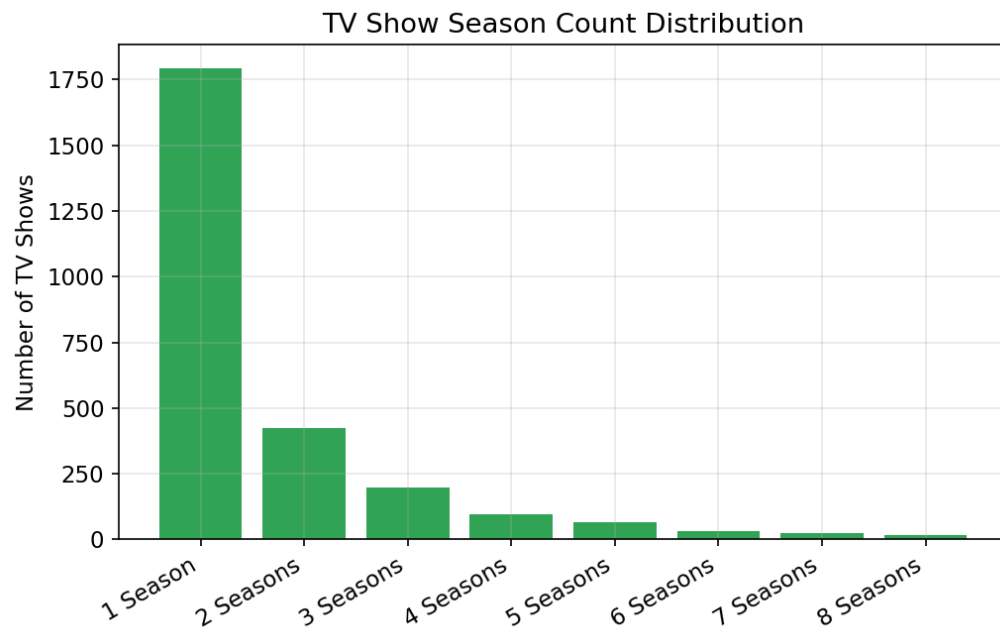


Figure 10. TV show season-count distribution

INSIGHTS:

- The large majority of TV shows on the platform run for just one season.
- Multi-season renewals become progressively rarer beyond season 2–3.
- Season-count patterns can help forecast content-refresh and renewal needs.

KPI:

Average seasons per show: 1.76

Share with exactly 1 season: 67%

13. Top Directors by Title Count

A director is listed for 70.1% of titles, spanning 4,528 unique names. The chart below ranks the ten directors with the most credited titles.

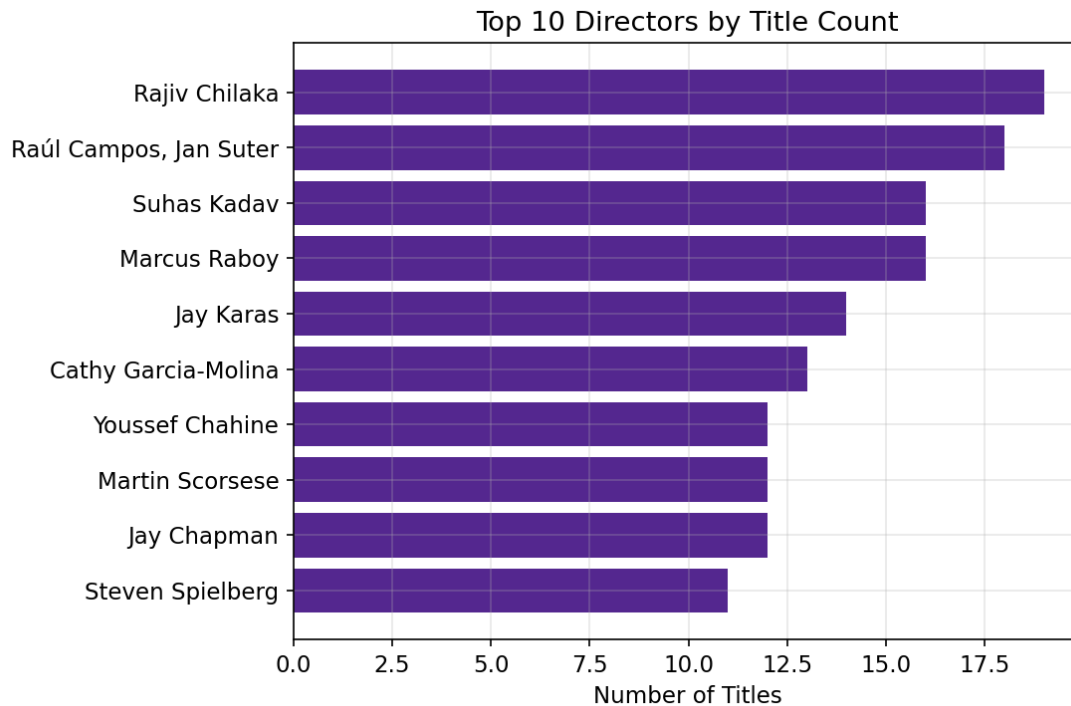


Figure 11. Top 10 directors by title count

INSIGHTS:

- Rajiv Chilaka tops the list with 19 credited titles in the supplied catalog.
- Several of the most prolific directors are associated with regional or serial content (e.g. animation, stand-up specials).
- Director data is missing for 29.91% of titles, so this ranking reflects only credited records.

KPI:

Most-credited director: Rajiv Chilaka

Titles credited: 19

Unique directors in catalog: 4,528

14. Genre Diversity vs. Title Count by Country

The relationship between a country's total title count and the number of distinct genres it covers provides an additional descriptive view of catalog breadth. Countries with more titles tend to also cover a wider range of genres, but the scatter plot should be read as an association rather than proof of causation.

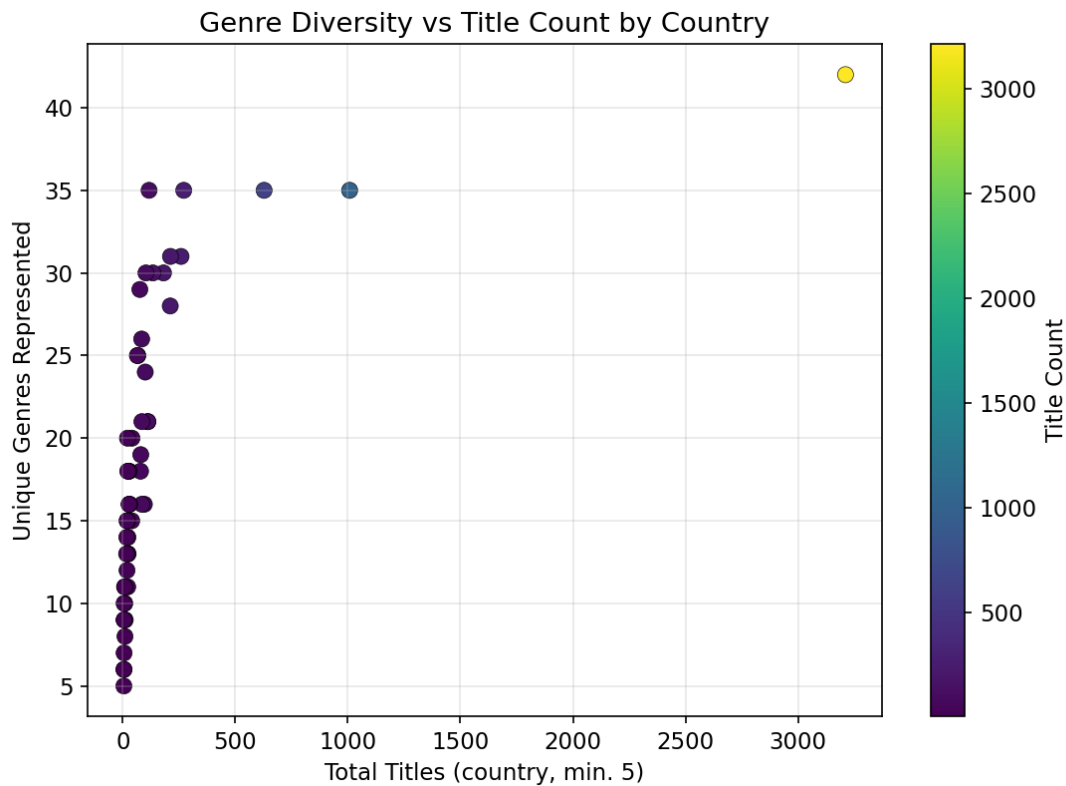


Figure 12. Genre diversity vs. title count by country (countries with ≥ 5 titles)

INSIGHTS:

- A positive association is visible: countries with larger catalogs also tend to cover more genres.
- This does not prove that a larger catalog causes greater genre diversity — both may reflect a country's overall content output.
- This view can support content-gap analysis for smaller or emerging markets.

KPI:

Countries plotted (≥ 5 titles): 54

Top country by title count: United States

Its genre diversity: 42 distinct genres

15. Content Additions by Day of Week

Breaking additions down by day of the week reveals whether new content tends to be released on particular days, which can be relevant for release-scheduling and marketing-cadence planning.

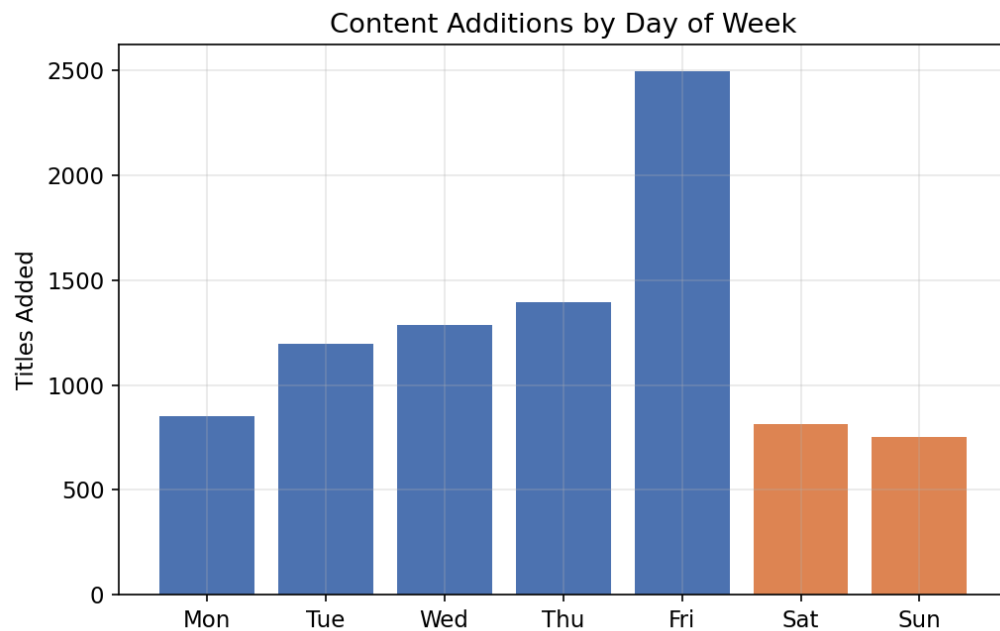


Figure 13. Content additions by day of week

INSIGHTS:

- Friday stands out as the most common day for new titles to be added, consistent with a weekly release cadence.
- Weekend days (Saturday and Sunday) see comparatively fewer additions than weekdays.
- This pattern can support scheduling marketing pushes around typical release days.

KPI:

Busiest addition day: Friday

Titles added on Fridays: 2,498

16. Weekday vs Weekend Addition Pattern

Across the row-level observations, average daily additions are 5.5 on weekdays versus 3.92 on weekends.

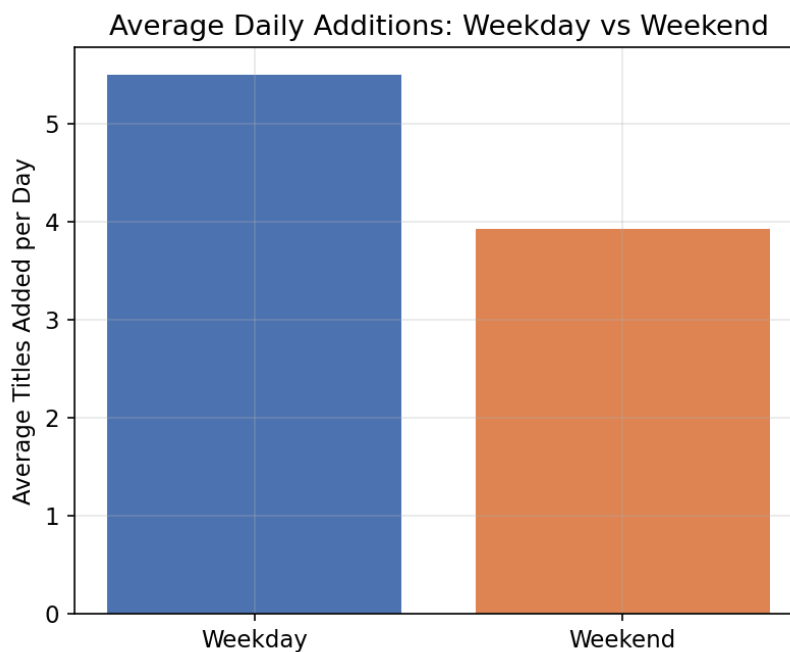


Figure 14. Average daily additions — weekday vs weekend

INSIGHTS:

- Weekday additions outpace weekend additions on average in the supplied data.
- The dataset is not a complete continuous daily series for every country/genre combination.
- The result should therefore be treated as a descriptive comparison of the available observations, not a guaranteed weekly cycle.

KPI:

Avg. additions per weekday: 5.5

Avg. additions per weekend day: 3.92

17. Content Freshness by Country

Content freshness — the gap between a title's release year and the year it was added to Netflix — averages 4.69 years across the supplied catalog, with 37% of titles added in their release year itself. Freshness is a useful catalog-quality indicator alongside raw title counts, similar in spirit to a customer-experience metric.

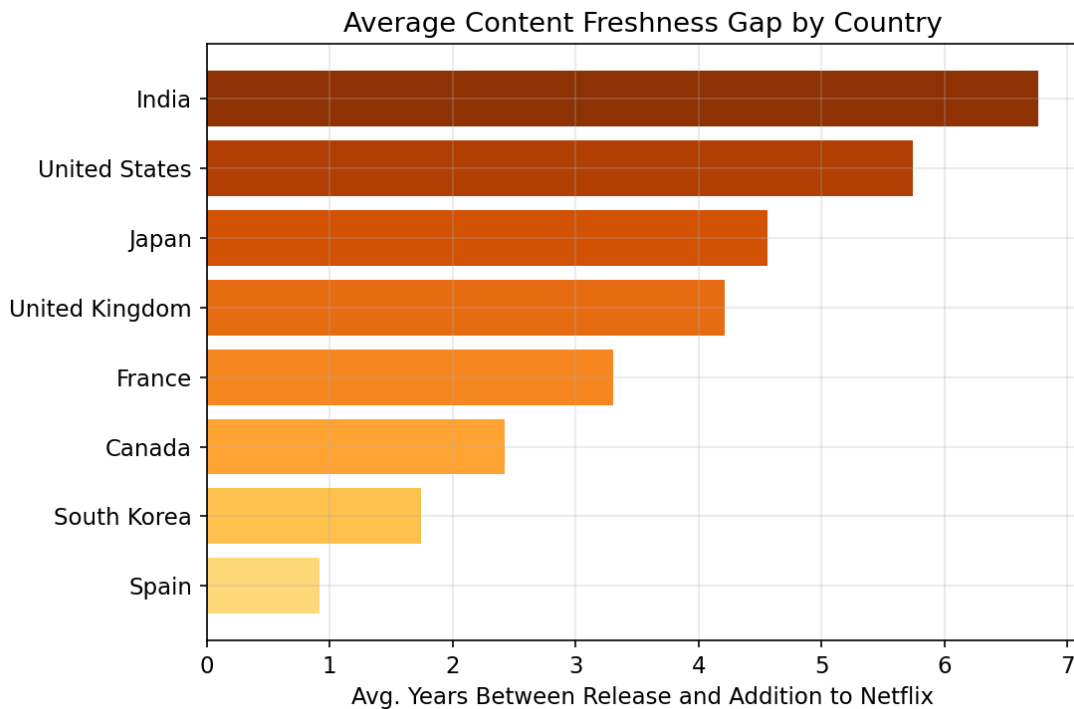


Figure 15. Average content freshness gap by country

INSIGHTS:

- Some countries' catalogs skew toward near-day-and-date releases, while others lean on older back-catalog licensing.
- A lower gap suggests titles from that country reach Netflix closer to their original release.
- Freshness should be considered alongside title volume to understand catalog quality more completely.

KPI:

Overall average freshness gap: 4.69 years

Share added in release year: 37%

18. Summary of Key Metrics

Metric	Value
Total titles	8,807
Movies / TV Shows	6,131 / 2,676
Date range (date added)	2008-01-01 to 2021-09-25
Unique countries referenced	127
Unique genre tags	42
Top genre	International Movies (2,752 titles)
Top country	United States (3,689 titles)
Most common rating	TV-MA (36.43%)
Average movie duration	99.6 minutes
Average TV seasons	1.76
Average content freshness gap	4.69 years
Dataset records	8,807

19. Next Steps to Move from Descriptive → Predictive

- Demand forecasting by genre and country to anticipate future catalog-growth needs.
- Anomaly detection for sudden shifts in addition rate, genre mix, or rating composition.
- Churn / engagement-impact analysis, linking catalog freshness to viewer retention (requires viewership data not present here).
- Content-gap analysis using genre-diversity-vs-title-count patterns to find under-served combinations.
- Trend and cohort analysis once richer, longer-running catalog snapshots become available.

This report answers what the supplied catalog snapshot shows. The next analytical layer would use viewership or engagement-level data, alongside a longer historical window, to explain why catalog patterns emerged and to forecast what is likely to happen next.

20. Glossary

I. Primary Country — the first country listed in a title's country field, used here as a simplified single-country attribution for titles credited to multiple countries.

II. Genre Tag — one of the (often multiple) category labels listed in the "listed_in" field for a title, e.g. "Dramas" or "International Movies".

III. Content Rating — the age/maturity classification assigned to a title (e.g. TV-MA, PG-13), broadly indicating intended audience.

IV. Content Freshness Gap — the number of years between a title's original release year and the year it was added to the Netflix catalog, as computed for this report.

V. Season Count — for TV shows, the number of seasons listed in the duration field; for movies, the duration field instead records runtime in minutes.